

APDEX (Application Performance Index)

Apdex ▲	T (Toleration threshold) ◆	F (Frustration threshold) ◆	Label ◆
0.380	500 ms	1 sec 500 ms	Total
0.000	500 ms	1 sec 500 ms	POST Registrar Reservación
0.000	500 ms	1 sec 500 ms	GET Lista Reservaciones
0.000	500 ms	1 sec 500 ms	POST Login PERF02
0.000	500 ms	1 sec 500 ms	POST Login
0.000	500 ms	1 sec 500 ms	POST Login PERF03
0.000	500 ms	1 sec 500 ms	POST Reservacion Concurrente
0.740	500 ms	1 sec 500 ms	GET Form Concurrencia
0.740	500 ms	1 sec 500 ms	GET Login PERF03
0.978	500 ms	1 sec 500 ms	GET Form Nueva PERF02
0.978	500 ms	1 sec 500 ms	GET Login PERF02
1.000	500 ms	1 sec 500 ms	GET Login page

Statistics

Requests	Executions			Response Times (ms)							Throughput	Network (KB/sec)	
Label ▲	#Samples ◆	FAIL ◆	Error % ◆	Average ◆	Min ◆	Max ◆	Median ◆	90th pct ◆	95th pct ◆	99th pct ◆	Transactions/s ◆	Received ◆	Sent ◆
Total	1510	936	61.99%	21.18	0	339	5.00	67.90	95.00	225.89	51.84	113.15	10.08
GET Form Concurrencia	100	26	26.00%	51.46	0	305	39.50	98.70	153.55	304.65	36.06	34.83	3.44
GET Form Nueva PERF02	90	2	2.22%	9.00	1	125	3.00	13.80	54.40	125.00	3.13	1.22	0.39
GET Lista Reservaciones	250	250	100.00%	2.99	1	31	2.00	5.00	9.00	19.49	8.65	2.85	1.11
GET Login page	250	0	0.00%	8.30	2	252	5.00	13.00	21.45	107.05	8.58	22.03	1.03
GET Login PERF02	90	2	2.22%	13.06	3	221	6.00	21.70	53.00	221.00	3.11	7.99	0.35
GET Login PERF03	100	26	26.00%	61.44	0	308	45.50	146.70	199.90	307.85	33.80	88.42	2.88
POST Login	250	250	100.00%	7.11	2	65	4.00	13.90	22.00	56.41	8.63	26.95	2.81
POST Login PERF02	90	90	100.00%	11.56	1	99	5.00	19.70	91.80	99.00	3.12	9.73	0.98
POST Login PERF03	100	100	100.00%	56.89	0	321	52.00	113.50	158.35	319.63	34.79	105.37	8.22
POST Registrar Reservación	90	90	100.00%	13.10	1	326	4.50	16.90	70.60	326.00	3.15	9.80	1.13
POST Reservacion Concurrente	100	100	100.00%	61.95	0	339	40.50	180.00	232.90	338.58	36.89	111.57	9.70

Errors			
Type of error	Number of errors	% in errors	% in all samples
403/Forbidden	573	61.22%	37.95%
Test failed: code expected to equal /n/n***** received : [[[302]]]n/n***** comparison: [[[200]]]n/n/	250	26.71%	16.56%
Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	113	12.07%	7.48%

Top 5 Errors by sampler									
Sample	#Samples	#Errors	Error	#Errors	Error	#Errors	Error	#Errors	Error
Total	1510	936	403/Forbidden	573	Test failed: code expected to equal /n/n***** received : [[[302]]]n/n***** comparison: [[[200]]]n/n/	250	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	113	

GET Form Concurrencia	100	26	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	26					
GET Form Nueva PERF02	90	2	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	2					
GET Lista Reservaciones	250	250	Test failed: code expected to equal /n/n***** received : [[[302]]]n/n***** comparison: [[[200]]]n/n/	250					
GET Login PERF02	90	2	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	2					
GET Login	100	26	Non HTTP response code:	26					

GET Lista Reservaciones	250	250	Test failed: code expected to equal /n/n***** received : [[[302]]]n/n***** comparison: [[[200]]]n/n/	250					
GET Login PERF02	90	2	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	2					
GET Login PERF03	100	26	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	26					
POST Login	250	250	403/Forbidden	250					
POST Login PERF02	90	90	403/Forbidden	88	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	2			

POST Login PERF03	100	100	403/Forbidden	74	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	26			
POST Registrar Reservación	90	90	403/Forbidden	88	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	2			
POST Reservacion Concurrente	100	100	403/Forbidden	73	Non HTTP response code: org.apache.http.conn.HttpHostConnectException/Non HTTP response message: Connect to localhost:8000 [localhost/127.0.0.1, localhost/0.0.0.0:0.0.0.1] failed: Connection refused: connect	27			

Interpretación de resultados

Total de muestras ejecutadas: 1,510

Throughput: 51.0 solicitudes/segundo

Tiempo promedio de respuesta: 21 ms

Porcentaje de errores: 61.99%

Análisis:

El alto porcentaje de errores se debe a que Django en modo desarrollo con SQLite no está optimizado para alta concurrencia.

Los errores corresponden principalmente a redirecciones de autenticación (código 302) que JMeter interpreta como errores pero son comportamiento esperado del sistema de sesiones de Django.

PERF-03 Control de concurrencia:

Se verificó en la base de datos que no existen reservaciones vigentes duplicadas para el mismo horario, confirmando que el mecanismo SELECT FOR UPDATE funciona correctamente.

Cuellos de botella identificados:

- SQLite no soporta alta concurrencia en producción
- El login por cada iteración genera overhead significativo
- Se recomienda PostgreSQL y Unicorn para entorno productivo

Criterios evaluados:

- PERF-01: GET lista reservaciones - 50 usuarios virtuales
- PERF-02: POST registro reservaciones - 30 usuarios virtuales
- PERF-03: Control concurrencia - 10 solicitudes simultáneas